

SHARIF UNIVERSITY OF TECHNOLOGY

Fault-Tolerant Deep Neural Network Hardware Accelerator Design

for Image and Video Processing Applications

M.Sc. THESIS · RESEARCH PROGRESS REPORT

Presented by **Ali Roshani**

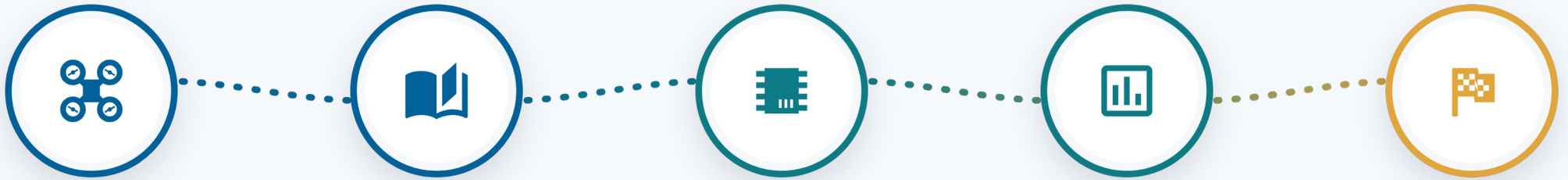
Supervisors **Dr. Mohsen Ansari** & **Dr. Mohammad T. Manzuri Shalmani**

Date **Aban 1405 · November 2026**

CPSLab Cyber-Physical Systems Laboratory

Outline

01 Introduction
02 Literature
03 Methodology
04 Results
05 Conclusion



01

Introduction

Mission-critical vision, the edge device, and why its neuron must not fail

Ch. 1–2

02

Literature Review

Six families of prior work and where this thesis stands

Ch. 3

03

Methodology

Hardened neuron → trustworthy fault analysis → BioVision

Ch. 4–7

04

Results

PPA, fault campaign, RFoM, on-chip learning, BioVision

Ch. 8

05

Conclusion

Takeaways, publications and the road ahead

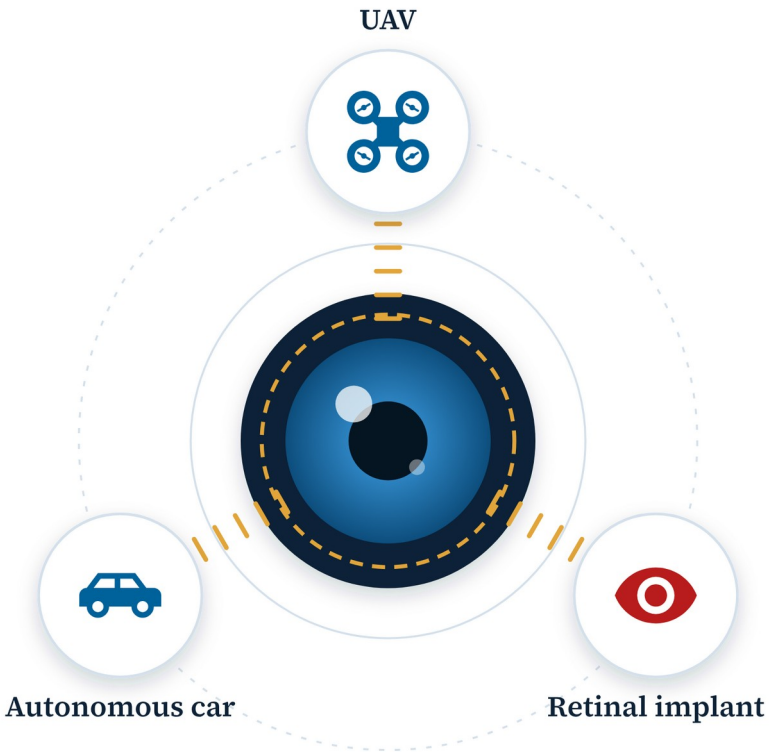
Ch. 9

THESIS STATEMENT

The system can share the load, but not the responsibility: **fault tolerance must move down the stack — from the system to the accelerator, and into the neuron.**

01

Introduction

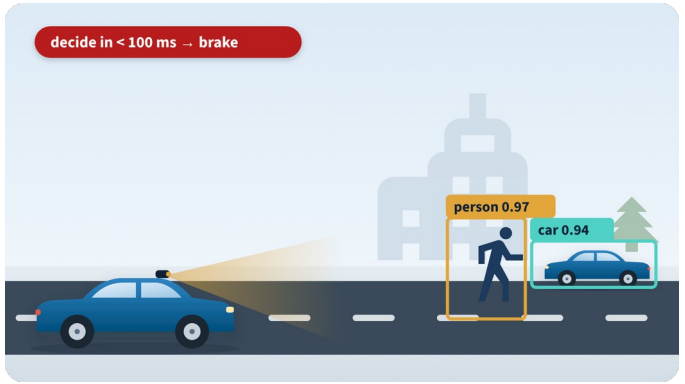


an image must become a **decision** — on the device itself

Our Thesis and Where It Matters

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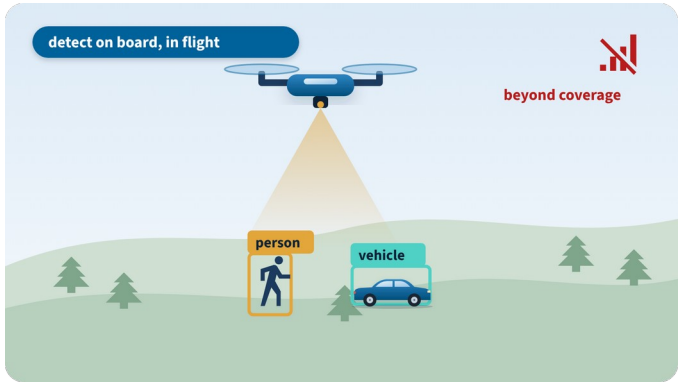
THIS THESIS A fault-tolerant hardware accelerator for image and video processing — **dependable vision on the device itself.**



Autonomous vehicles

On-board perception must decide within milliseconds.

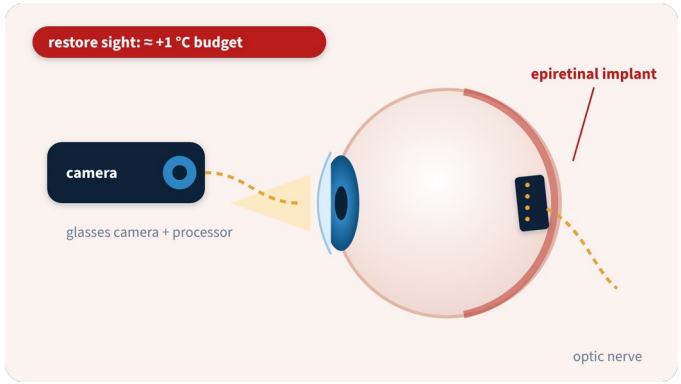
A missed obstacle is a safety failure.



UAVs and robots

Detection must keep working in flight, beyond coverage.

No repair until the mission ends.



Retinal implants

Artificial sight for the blind runs inside the body.

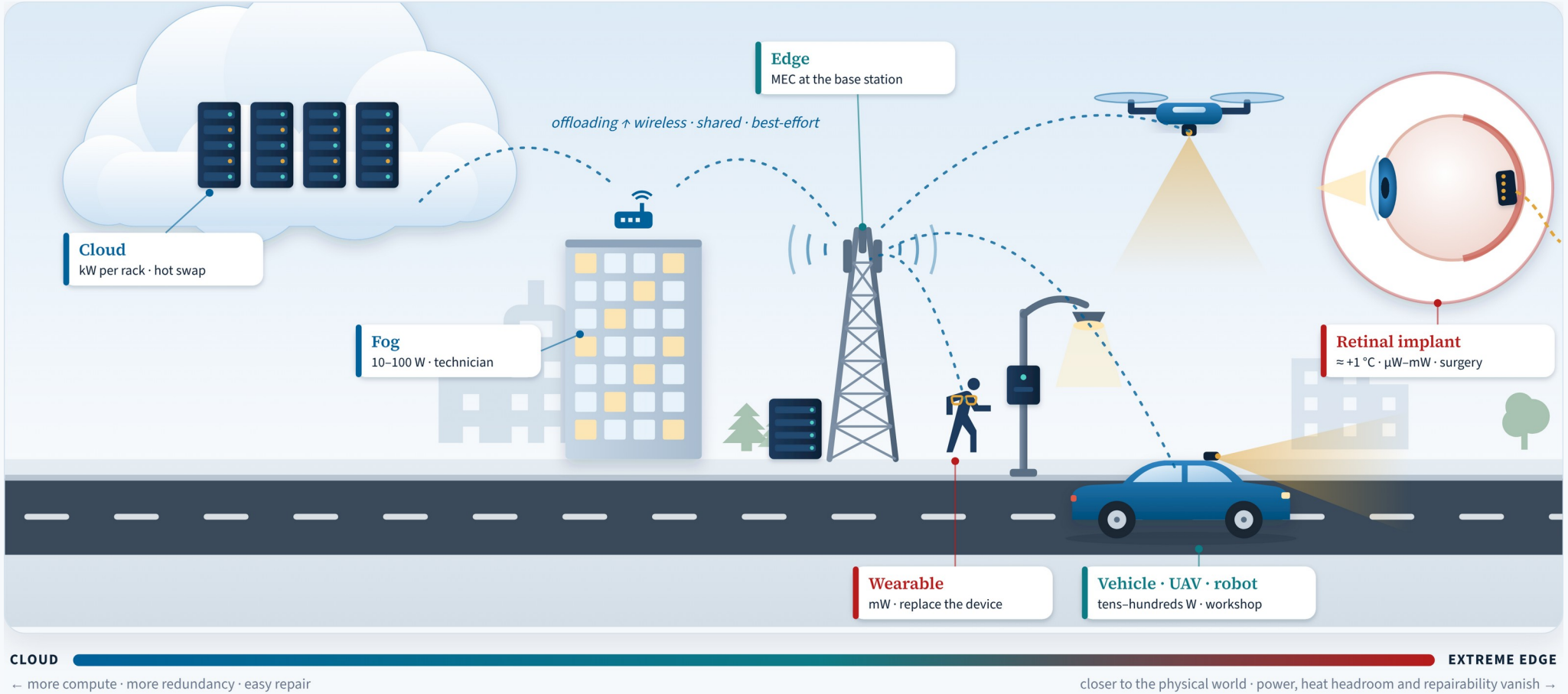
≈ +1 °C budget; repair means surgery.

Common need: **real-time · low-power · dependable** image processing at the extreme edge

Safety: ISO 26262 (road vehicles) · Implant thermal limit ≈ +1 °C: Kim et al., IEEE TNSRE 2007 · Retinal prostheses: refs. [62, 63] of Ch. 1

Where Should the Computation Live?

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The systems answer was offloading — it quietly assumes **the end device is a correct black box**

Cloudlets: Satyanarayanan 2009 · Fog: Bonomi 2012 · Edge: Shi 2016 · Platform power: Capra et al. 2020 · Implant limit: Kim et al. 2007 (thesis Table 1.1)

Why Offloading Fails Safety-Critical Vision

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01 Time & determinism

Latency becomes a distribution: 1 % slow servers make 63 % of requests slow.

02 Connectivity

Coverage gaps, jamming and tissue attenuation cut the link when it is needed most.

03 Energy & heat

Sending 1 kbit over 100 m costs as much energy as 3 M instructions.

04 Security & privacy

Every hop is an attack surface; split learning leaks private data.

05 Dependability & certification

An open, shared chain cannot be certified to IEC 61508 / ISO 26262.

06 Silent data corruption

Even cloud cores corrupt data silently — a few per several thousand machines.

07 Scale & economics

Spectrum and fog / MEC infrastructure are finite and costly to run.

08 Environment & modelling

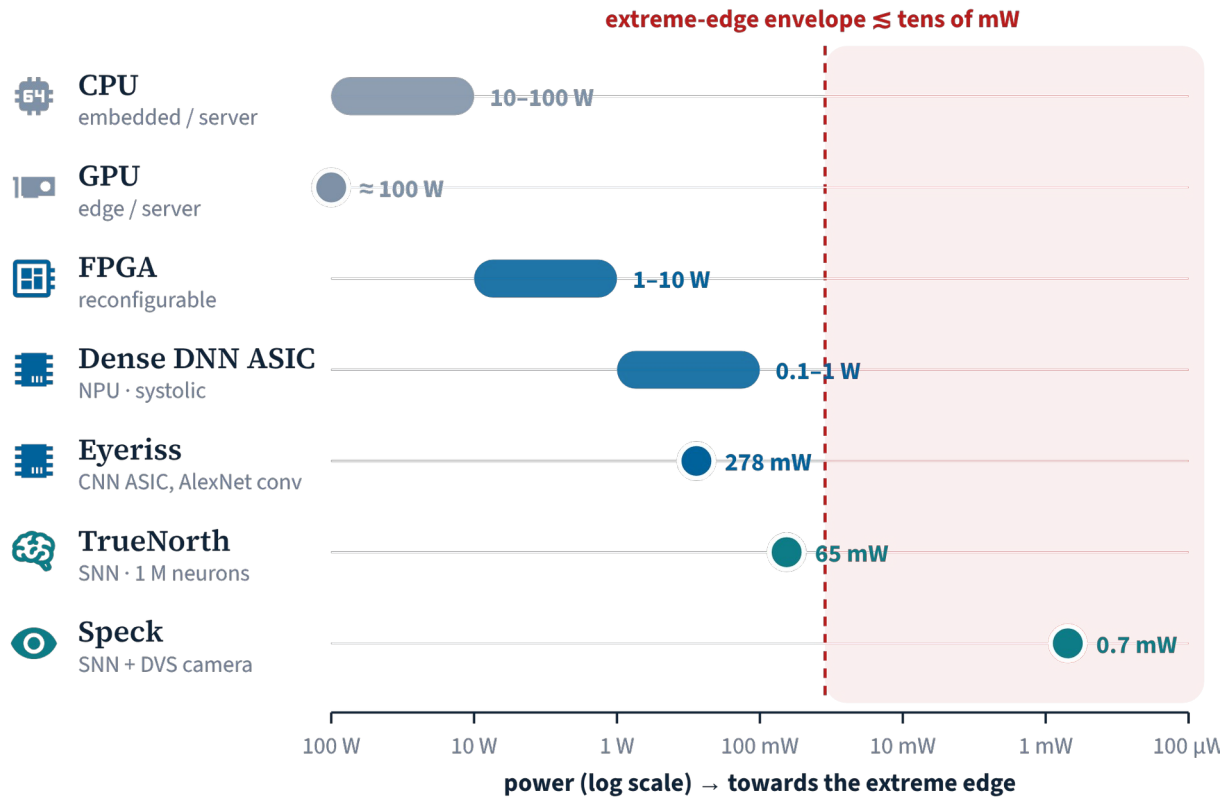
Radiation, heat and vibration hit the device anyway; offloading models assume it is correct.

8 weakness groups → **the critical perception–decision loop must close on the device itself**

Dean & Barroso 2013 · Pottie & Kaiser 2000 · Pasquini et al. 2021 · Koopman & Wagner 2016 · Hochschild et al. 2021 · Dixit et al. 2021 · Baumann 2005

Which Accelerator for the Extreme Edge?

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Dense NPU / systolic ASIC

Efficient for CNNs — but computes every pixel, every frame.

QNN · TinyML

Quantisation and pruning shrink the model, not the dense computation.

✓ OUR CHOICE

SNN · neuromorphic

Computes only on spikes — event-driven, sparse, mW-scale.

So we build our own extreme-edge device on SNN — **local · real-time · within power and heat · accurate**

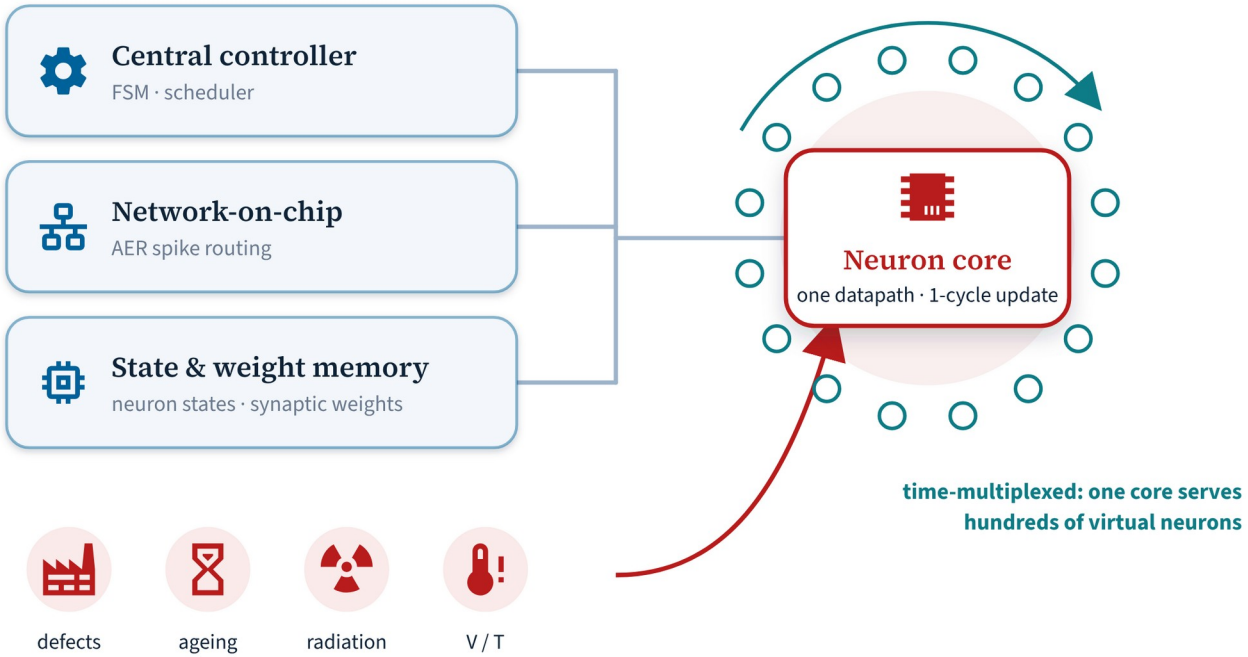
Capra et al. 2020 (platform power) · Chen et al. 2017 (Eyeriss) · Akopyan et al. 2015 (TrueNorth) · Yao et al. 2024 (Speck) · implant envelope: Kim et al. 2007

Why Dependability Must Be Built In

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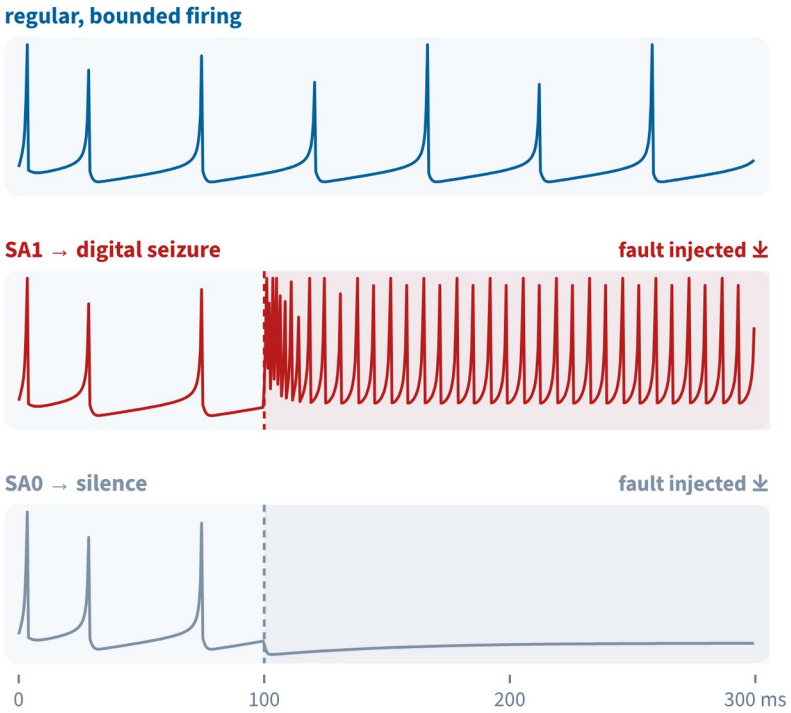
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Time-multiplexed SNN accelerator (ODIN-style)



permanent faults → a node stuck-at-0 / stuck-at-1 inside the shared core

What one stuck-at does to a neuron



Izhikevich RS neuron (behavioural model): SA1 ≈ drive stuck high, SA0 ≈ drive lost

Alone at the edge, nothing can outvote a faulty core → **harden the shared neuron core first**

Frenkel et al. 2019 (ODIN) · Izhikevich 2003 · Baumann 2005 · Vatajelu et al. 2019 · Spyrou et al. 2021 · Hochschild et al. 2021

Research Gaps and Contributions

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THE GAP	OUR ANSWER	
G1 No deterministic guarantees deadline · power · energy · accuracy met only statistically	C1 · H-ASIZH neuron core	*Phase 1* Deterministic by design: one-cycle update, Lyapunov-bounded, 182.8 µW for two cores.
G2 Standards rarely met e.g. ISO 14708: ≤ 2 °C implant heating	C2 · GateScan	*Phase 2* Certified fault brackets instead of percentages — evidence a standard can accept.
G3 The device is assumed correct reliability, availability, security barely studied	C1 · H-ASIZH neuron core	*Phase 1* A self-protecting core: more than 4× better than full TMR, with one-third less silicon.
G4 Neuromorphic is not mature still below YOLO-class accuracy on COCO	C3 · BioVision	*Phase 3* Brain-like spiking vision with local learning: classification done, COCO detection next.
G5 No learning in the field devices drift and wear; learning stops	C4 · On-chip learning	*Phase 1* STDP learning on the hardened core: 1020 / 1024 neurons stay active, 81.8 % on MNIST.

Out of scope: security attacks and rare long-tail events.

Buttazzo, Hard Real-Time Computing Systems · ISO 14708-1:2014 · ISO 26262:2018 · Roshani et al., TCAS-I (submitted) · Roshani et al., GateScan, TCAD (draft)

02

Literature Review



Where This Thesis Stands

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Family	Work	Edge vision	Hardware		Accelerator type	Fault tolerance		Fault-assessment campaign	On-chip learning
			ASIC	FPGA		Permanent	Transient		
A Edge-vision accelerators	Eyeriss · JSSC 2017	✓	✓ 65 nm	×	CNN PE array	×	×	×	×
	Suh et al. (UAV) · TRETS 2023	✓	×	✓	CNN engine	×	×	×	×
B Fault-tolerant accelerators	Zhang et al. · VTS 2018	×	◐ model	×	Systolic array	✓ stuck-at	×	behavioural simulation	×
	YOLO on Versal (BYU) · 2026	◐ space	×	✓	CNN engine	×	✓ SEU	config-memory injection + proton beam	×
C Neuromorphic processors	ODIN · TBioCAS 2019	◐ MNIST	✓ 28 nm	×	Neuromorphic NPU	×	×	×	✓
	Speck · Nat. Commun. 2024	✓ DVS	✓	×	Neuromorphic NPU	×	×	×	×
	Loihi 2 · 2021	×	✓ Intel 4	×	Neuromorphic NPU	×	×	×	✓
D SNN fault tolerance	Spyrou et al. · DATE 2021	◐ N-MNIST	×	×	SW model	✓ dead / sat.	×	behavioural	×
	SoftSNN · DAC 2022	◐ MNIST	✓ 65 nm	×	Neuromorphic NPU	×	✓ soft err.	behavioural	×
	RescueSNN · Front. Neuro. 2023	◐ MNIST	✓ 65 nm	×	Neuromorphic NPU	✓ stuck-at	×	behavioural	×
E Fault-injection tools	ShadowFI · IEEE Access 2025	×	✓ netlists	✓ emul.	GPU units	—	—	gate-level · 2-valued · SDC / DUE / masked	×
	SpikeFI · 2024	×	×	×	GPU (SW)	—	—	software (PyTorch)	×
F Spiking vision systems	Kheradpisheh et al. · 2018	×	×	×	CPU / GPU (SW)	×	×	×	◐ STDP, SW
	SpikeYOLO · ECCV 2024	×	×	×	GPU (SW)	×	×	×	×
★	**This thesis**	✓ µW cores	✓ Sky130	×	**Neuromorphic NPU**	✓ all nodes	◐ assessed	**gate-level · 4-valued · certified bracket**	✓ STDP

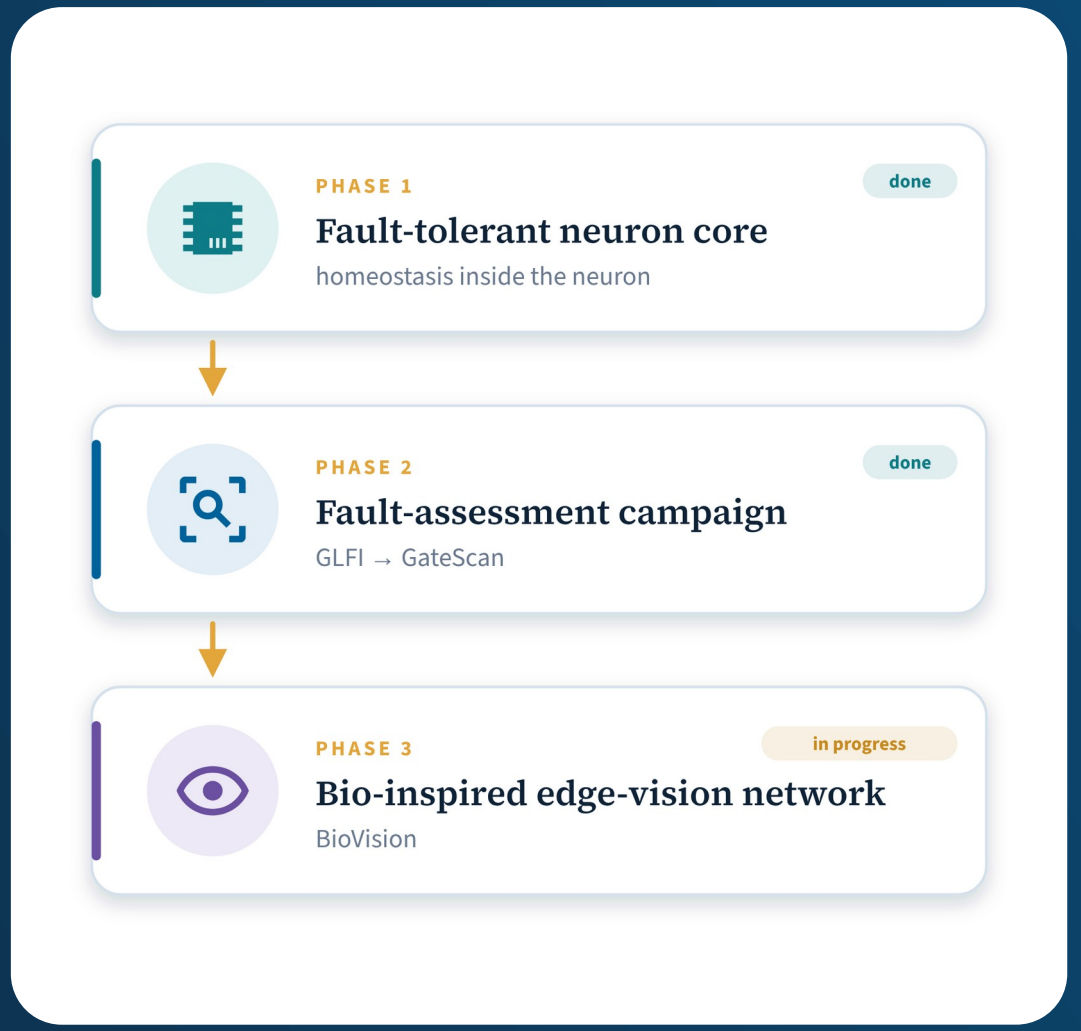
✓ yes ◐ partial / model / assessed only × no Edge vision = on a low-power edge device, not a GPU / PC · NPU = neural processing unit (neuromorphic) · systolic / PE array = dense DNN accelerator

Only this thesis combines **edge vision · permanent-fault tolerance · a faithful campaign · on-chip learning**

Chen 2017 · Suh 2023 · Zhang 2018 · BYU 2026 · Frenkel 2019 · Yao 2024 · Davies 2021 · Spyrou 2021 · Putra 2022, 2023 · Guerrero-Balaguera 2025 · Spyrou 2024 · Kheradpisheh 2018 · Luo 2024

03

Methodology



Our Path and Our Toolbox

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PHASE 1

Fault-tolerant neuron core

Of the four vulnerable parts of an ODIN-style accelerator, we harden the Izhikevich / LIF neuron first — with Lyapunov-guided homeostasis, not redundancy.

Bonus: the adaptive threshold is also a learning rule — overhead ≈ only α, β, γ.

PHASE 2

Fault-assessment campaign

We build GLFI, grow it into GateScan, and test every node of the core with it.

PHASE 3

Bio-inspired edge-vision network

We scale the hardened core into a network and build BioVision on it.

RTL → GDSII · FULLY OPEN-SOURCE, ASIC-TARGETED FLOW

RTL design
(VHDL / Verilog)
- Fault-tolerant neuromorphic processor
- Homeostatic feedback loops
- Error detection & recovery

STAGE 1 LOGIC SYNTHESIS — Yosys *open-source synthesis*
- RTL parsing & elaboration
- Boolean optimization
- Redundancy extraction
- Technology mapping
- Timing & area optimization

SkyWater 130 nm standard-cell library

STAGE 2 PHYSICAL IMPLEMENTATION (P&R) — OpenROAD
- Floorplanning
- Power grid generation (PDN)
- Placement (global + detailed)
- Clock tree synthesis (CTS)
- Routing (global + detailed)
- Filler, tap cells & DRC fixing

Handles the routing density of the homeostatic feedback loops

STAGE 3 SIGN-OFF & ANALYSIS — OpenSTA
- Post-synthesis STA
- Multi-corner timing analysis
- Critical path & slack analysis
- Power analysis (PPA)
- DRC / LVS (via Magic)
- GDSII stream-out

Power: leakage · internal · switching
Area: cell + routing · Fmax · worst slack

OUTPUT · TAPE-OUT READY
- **GDSII** layout
- **Sign-off reports** STA · PPA · DRC
- **Netlist** post-route

VERIFICATION & FAULT-INJECTION FRONTEND · PRE-SYNTHESIS

RTL source code (VHDL / Verilog) → GHDL (VHDL compiler & simulator) → cocotb (Python testbenches · VPI / VHPI) → Cycle-accurate simulation (4-valued logic (0 · 1 · X · Z)) → Gate-level fault injection (GLFI campaigns → GateScan) → Coverage & metrics (outcome classes · debug)

new workloads / triggers

DEVELOPMENT ENVIRONMENT & TOOLING

- **Linux on WSL 2** — Yosys & netlist tools on Windows
- **Visual Studio Code** — RTL development & debugging
- **Docker** — reproducible EDA environment
- **Open-source EDA stack** — Yosys · OpenROAD · OpenSTA · Magic · GHDL · cocotb · Icarus
- **SkyWater 130 nm PDK** — open, commercial-grade CMOS

Toolbox: Yosys · OpenROAD · OpenSTA · Magic · GHDL · cocotb · Icarus Verilog · SkyWater 130 nm PDK · WSL 2 · Docker · VS Code

Phase 1 · Starting Point: the Izhikevich Neuron

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IZHIKEVICH MODEL (2003) · RICH FIRING FROM FOUR PARAMETERS a, b, c, d

$$\frac{dv}{dt} = 0.04\,v^2 + 5v + 140 - u + I$$
$$\frac{du}{dt} = a\,(b\,v - u)$$
$$v \geq 30\ \text{mV} \;\Rightarrow\; v \leftarrow c, \quad u \leftarrow u + d$$

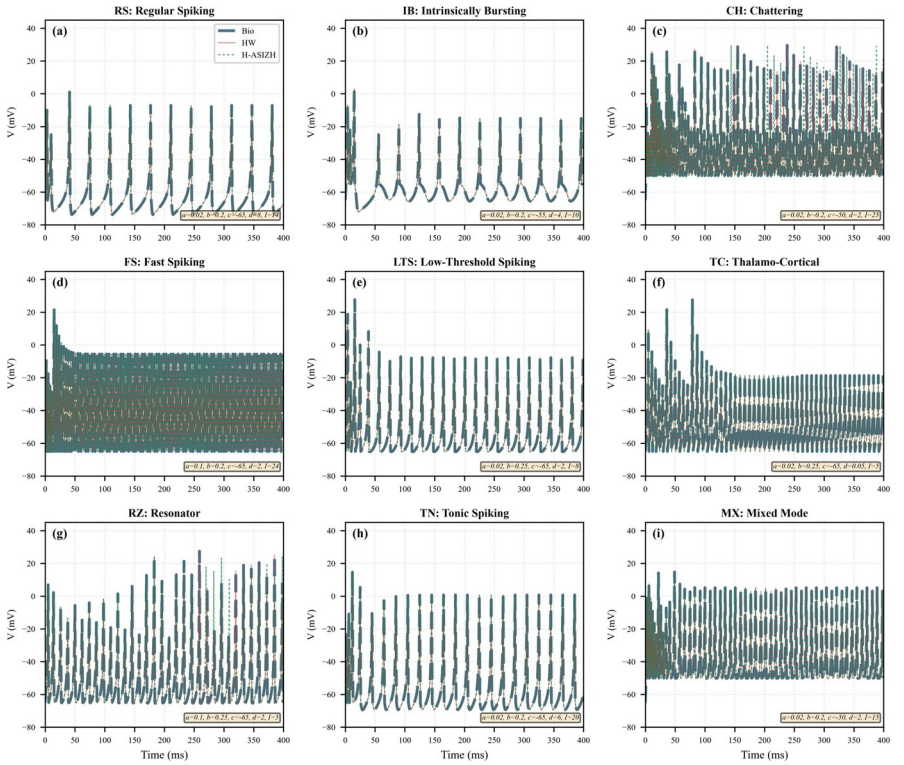
EXACT AFFINE MAP · ÷64 IS A FREE RIGHT-SHIFT

$$V = \frac{v + 65}{64}, \qquad U = \frac{u}{64}, \qquad I_{hw} = \frac{I}{64}$$

HARDWARE FORM · SAME NULLCLINES, SAME PHASE PORTRAIT

$$\frac{dV}{dt} = 2.56\,V^2 - 0.2\,V + I_{hw} - U - 0.25$$
$$\frac{dU}{dt} = a\left(b\,V - U - \frac{65b}{64}\right)$$

Contraction keeps the state near [−0.5, 1.5]: **free guard bits** in the Q7.19 word.



Bio vs hardware model: the nine firing patterns overlap

Exact and shift-friendly — **yet one stuck-at still drives it into a digital seizure**

Izhikevich, IEEE TNN 2003 · Roshani et al., H-ASIZH, IEEE TCAS-I (submitted), Sec. II

Adding Homeostasis, Guided by Lyapunov

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H-ASIZH · THE THRESHOLD θ BECOMES A SLOW STATE VARIABLE

$$\frac{dV}{dt} = 2.56\,V^2 - 0.2\,V + I_{hw} - U - 0.25$$

$$\frac{dU}{dt} = a\left(b\,V - U - \frac{65b}{64}\right)$$

$$\frac{d\theta}{dt} = \alpha\,(\theta_0 - \theta) + \gamma \sum_k \delta(t - t_k)$$

$$V \geq \theta \;\Rightarrow\; V \leftarrow c_{hw},\;\; U \leftarrow U + d_{hw} + \beta\,(\theta - \theta_0)$$

$\alpha = 2^{-6}$ leak · $\gg 6$

θ returns to θ_0 when healthy

$\gamma = 2^{-3}$ penalty · $\gg 3$

every spike raises θ

$\beta = 2^{3}$ brake · $\ll 3$

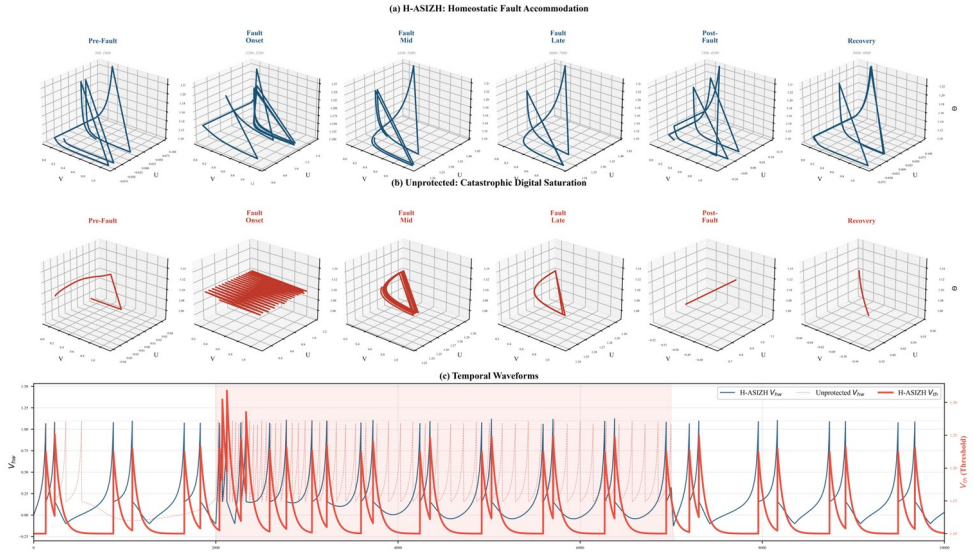
deeper reset after a burst

LYAPUNOV ARGUMENT

$$\mathcal{V}(\theta) = \tfrac{1}{2}\,(\theta - \theta^*)^2$$

$$\dot{\mathcal{V}} = (\theta - \theta^*)\big[-\alpha(\theta - \theta^*) + \gamma\,(r - r^*)\big] < 0$$

$$\Rightarrow\; |\theta - \theta^*| \lesssim \gamma \quad \text{(UUB)}$$



Normal: 2-D cycle · Fault: collapse · H-ASIZH: bounded 3-D attractor

The firing rate is **uniformly ultimately bounded** — the band is set by γ, not by the size of the fault.

θ makes the neuron its own controller — **no external monitor, no redundancy**

Khalil, Nonlinear Systems · Slotine & Li, Applied Nonlinear Control · Roshani et al., TCAS-I (submitted), Sec. III

The Same Equation Is a Learning Rule

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OUR FAULT-TOLERANCE CONTROLLER (H-ASIZH)

$$\frac{d\theta}{dt} = \underbrace{\alpha\,(\theta_0 - \theta)}_{\text{leak}} + \underbrace{\gamma \sum_k \delta(t - t_k)}_{\text{per-spike step}}, \qquad \text{spike if } V \geq \theta$$

≡

ADAPTIVE-THRESHOLD HOMEOSTASIS USED FOR UNSUPERVISED STDP LEARNING (DIEHL & COOK 2015)

$$\frac{d\theta_e}{dt} = \underbrace{\frac{1}{\tau_\theta}\,(v_{th} - \theta_e)}_{\text{leak}} + \underbrace{\theta^+ \sum_k \delta(t - t_k)}_{\text{per-spike step}}, \qquad \text{spike if } v \geq \theta_e$$

$$\alpha \leftrightarrow \frac{1}{\tau_\theta} \qquad \theta_0 \leftrightarrow v_{th} \qquad \gamma \leftrightarrow \theta^+$$

Biology

α ↔ Ca^{2+} clearance · γ ↔ Ca^{2+} influx per spike · β ↔ Ca^{2+}-dependent hyperpolarization

Hardware

Controller = three shifters, two adders, one comparator — no multiplier.

Learning

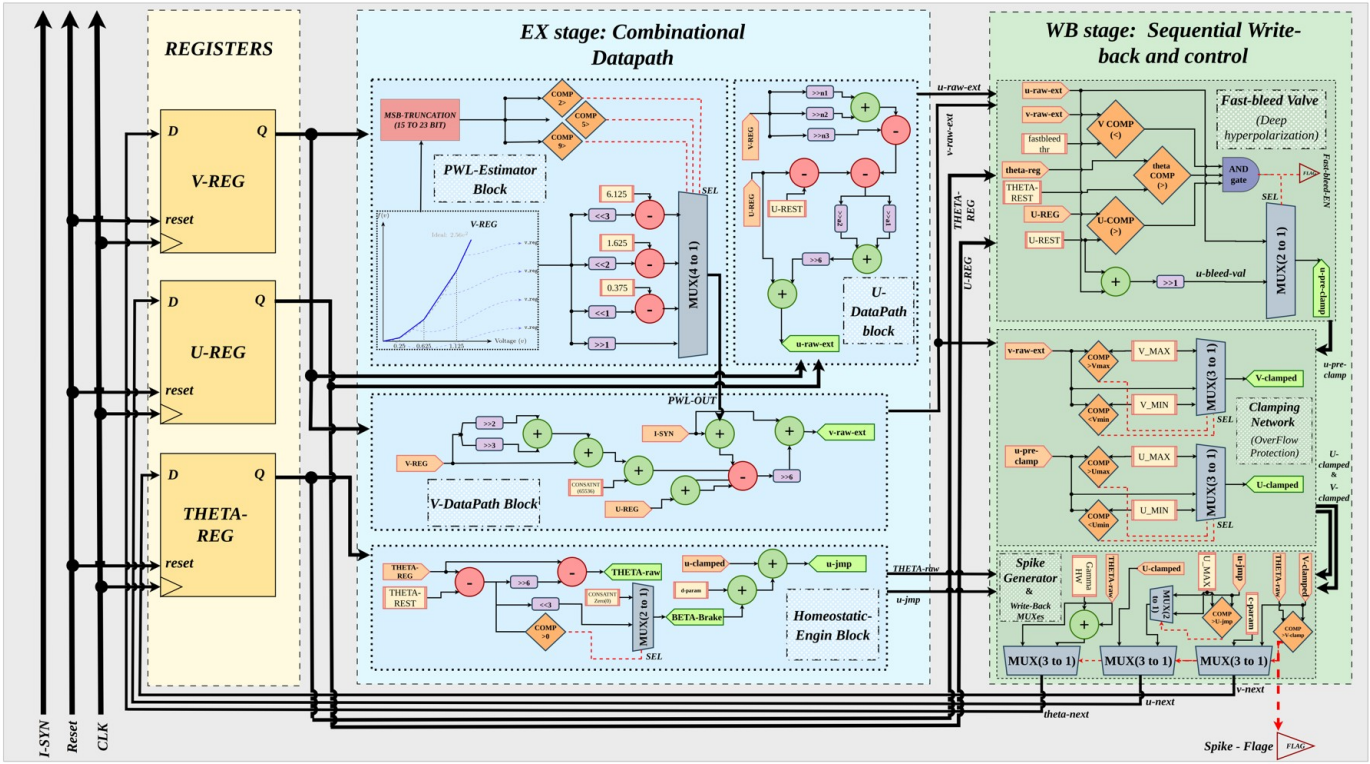
The same θ prevents neuron death and dominance in competitive learning.

Silicon bought for fault tolerance also buys on-chip learning — **overhead ≈ α, β, γ**

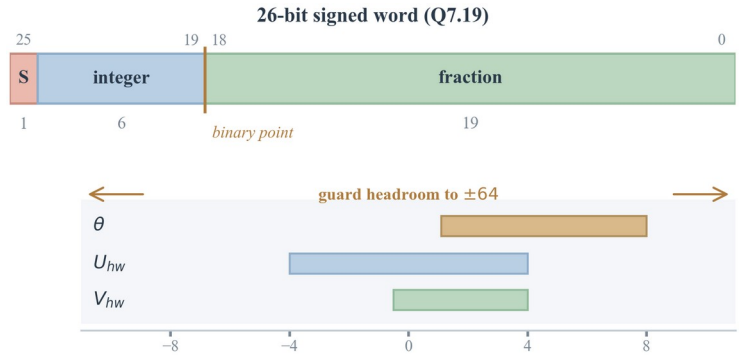
Diehl & Cook, Front. Comput. Neurosci. 2015 (adaptive threshold for STDP learning) · Johnson et al., TCAS-I 2018 · Roshani et al., TCAS-I (submitted)

A One-Cycle, Multiplier-Free Micro-Architecture

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EX: PWL estimator · V and U datapaths · homeostatic engine WB: clamping · fast-bleed valve · spike generator



V² WITHOUT A MULTIPLIER · 4-SEGMENT PWL

$$\mathrm{PWL}(V) = \begin{cases} 0 & V < 0 \\ V \gg 1 & 0 \le V < 0.25 \\ (V \ll 1) - 0.375 & 0.25 \le V < 0.625 \\ (V \ll 2) - 1.625 & 0.625 \le V < 1.125 \\ (V \ll 3) - 6.125 & V \ge 1.125 \end{cases} \approx 2.56\,V^2$$

1 update / cycle no latency jitter in a timing code

shifts & adds only no multiplier on the critical path

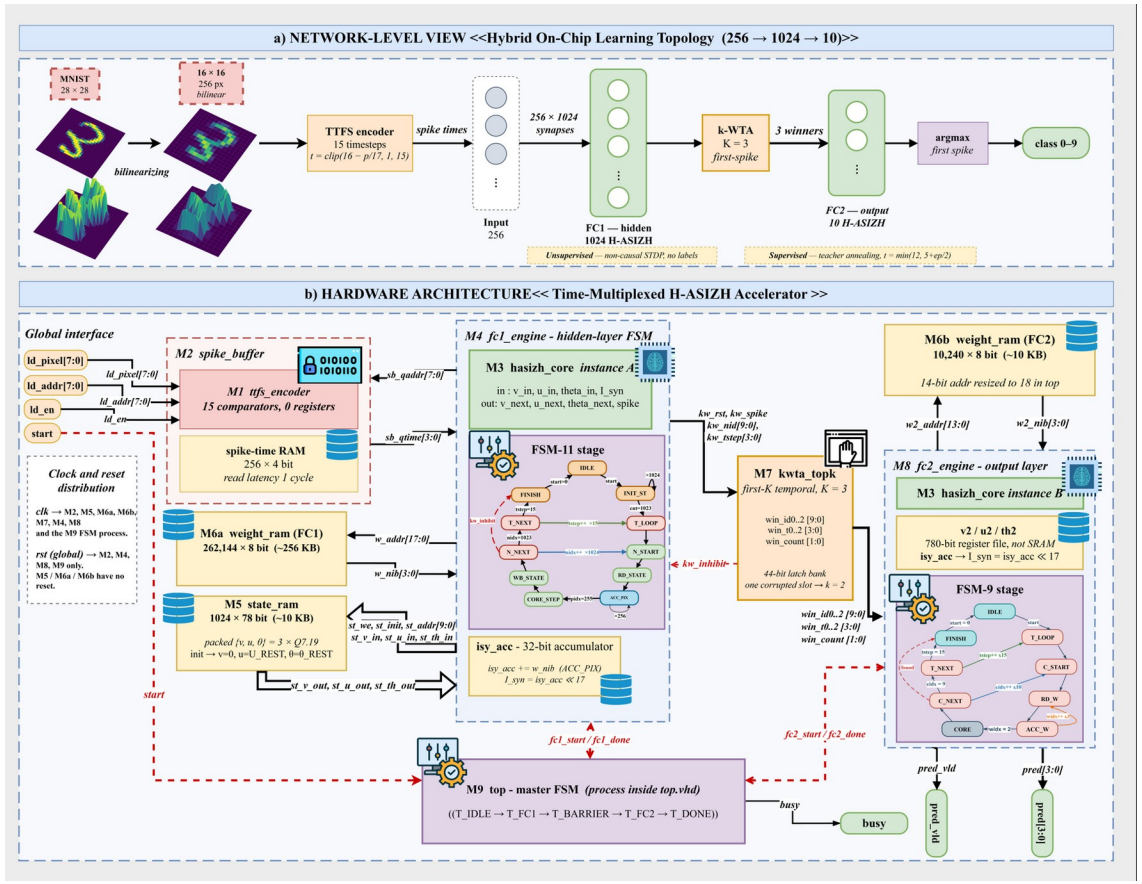
safety inside the core clamping · fast-bleed · β-brake

Placed and routed in SkyWater 130 nm — **the controller adds only shifters, adders and a comparator**

Roshani et al., TCAS-I (submitted), Sec. IV, Figs. 5–6 · RTL: GLFI_fault_campaign/rtl/mark2_homeostatic.vhd

Case Study: Does Learning Survive on the Core?

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1 Network

256 → 1024 → 10, TTFS-coded, built only from our neuron cores.

2 On-chip learning

Hidden layer: unsupervised STDP with first-K winner-take-all (K = 3).

3 Read-out

Output layer: delta rule with teacher annealing; the first spike decides the class.

4 Hardware

Two physical cores serve all logical neurons (time-multiplexed), with weight and state RAMs under one master FSM.

One question for the results: **does learning survive on the fault-tolerant core?**

Roshani et al., TCAS-I (submitted), Sec. VI, Fig. 8 · MNIST downsampled to 16 × 16, TTFS in 15 time steps

Phase 2 · How Do We Assess the Core?

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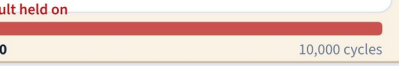
Phase 1 gave a core that rides out a parasitic current in simulation. **Now: how do we assess it — fault by fault, on the real netlist?**

F · Faults
where and what

- **site** — every extracted net of the netlist
- **value** — stuck-at-0 / stuck-at-1, permanent, single

A · Activation
under what and when

- **workload** — short, steady tonic-firing stimulus
- **time τ** — inject after 2,000 fault-free cycles, hold to 10,000



synthesized netlist

R · Readouts
what we observe

- **output** — spike timestamps — the only observation point
- **golden** — the same run without the fault

M · Measures
how we judge

- **band** — firing-rate error ≤ 20 % and ISI CV ≤ 0.20
- **classes** — masked · tolerant · critical · fatal

Census, not sampling

LEVEUGLE SAMPLE SIZE

$$n = \frac{N}{1 + e^2 \dfrac{N-1}{t^2\, p\,(1-p)}}$$

One core has only a few thousand faults, so the sample is already most of N (e.g. N = 5,000 → n ≈ 3,300 at e = 1 %, 95 %).

So we injected every site, in both polarities — **no fault collapsing, no fault dropping.**

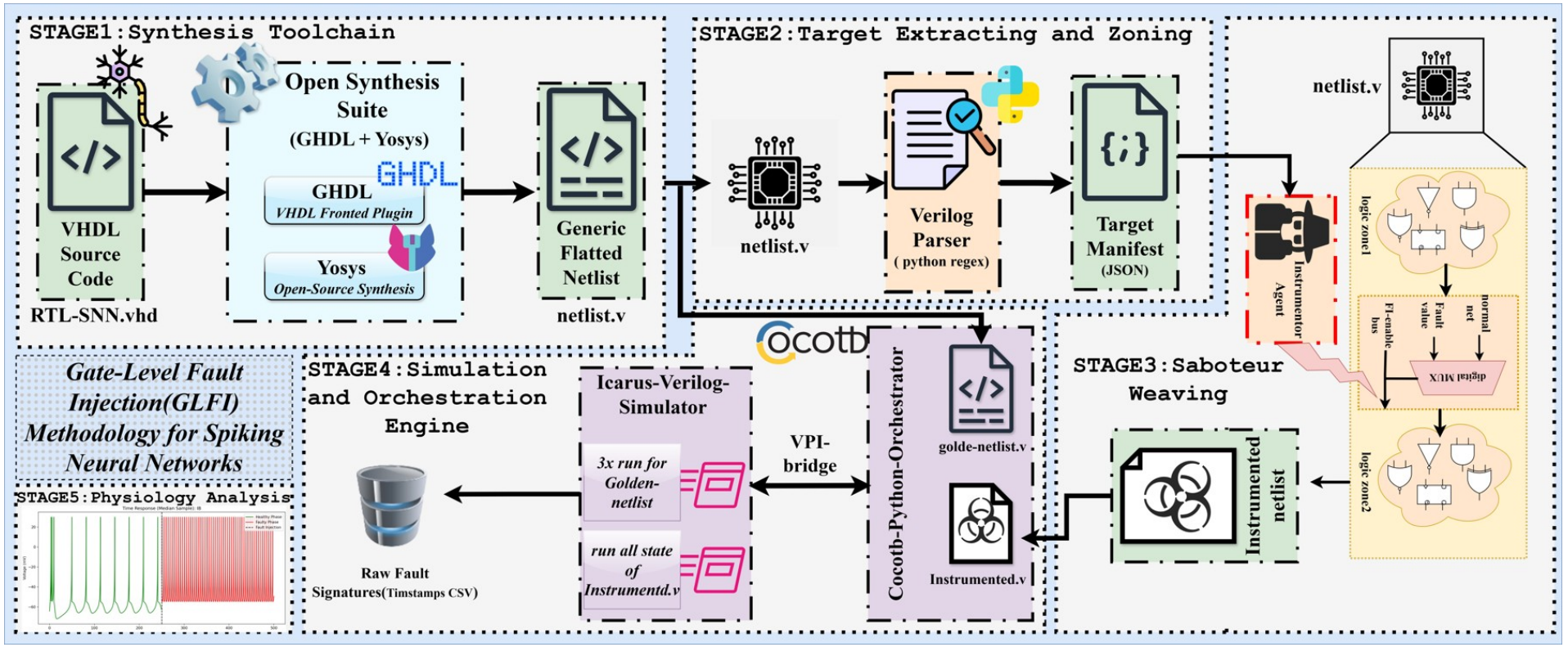
Netlist · site and value · workload · injection time — **every node, both polarities**

Arlat et al., IEEE TSE 1990 (FARM) · Leveugle et al., DATE 2009 · Roshani et al., TCAS-I (submitted), Sec. V

The GLFI Engine: Faults Woven Into the Netlist

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```
net = (fi_en && id==ID) ? fi_val : hook
```

Shadow Hook: one instrumented netlist serves the whole campaign

4-valued Icarus: X and Z propagate — no optimistic branch

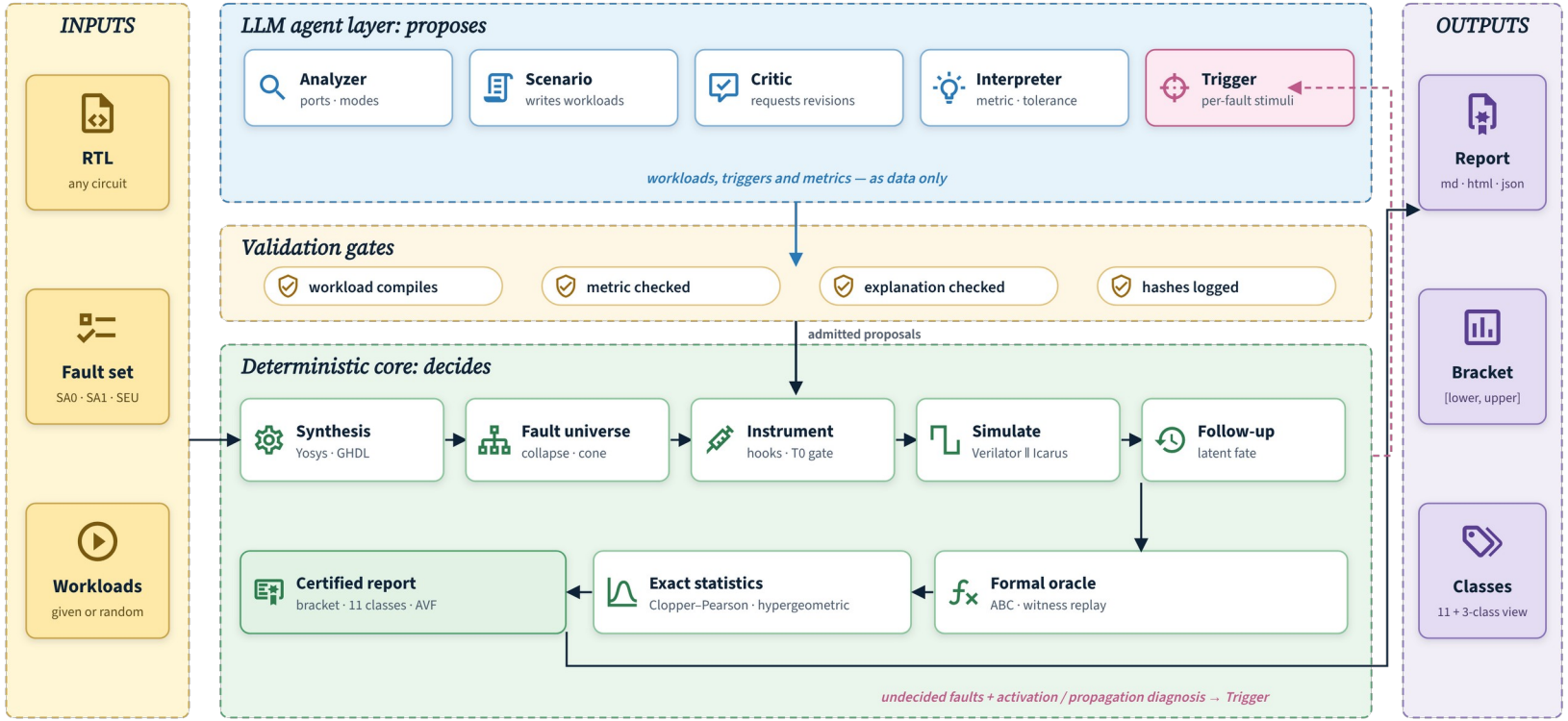
Neuro-metric: masked · tolerant · critical · fatal

Faults are compiled into the circuit — **not forced by the simulator**

Roshani et al., TCAS-I (submitted), Sec. V, Fig. 7 · github.com/aliroshanidino/GLFI_fault_campaign

Spin-off: From GLFI to GateScan

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Any circuit in

GLFI grew into a general tool: give it RTL, it returns a certified report.

Agents only propose

Five LLM roles write workloads, triggers and metrics — as data.

The core decides

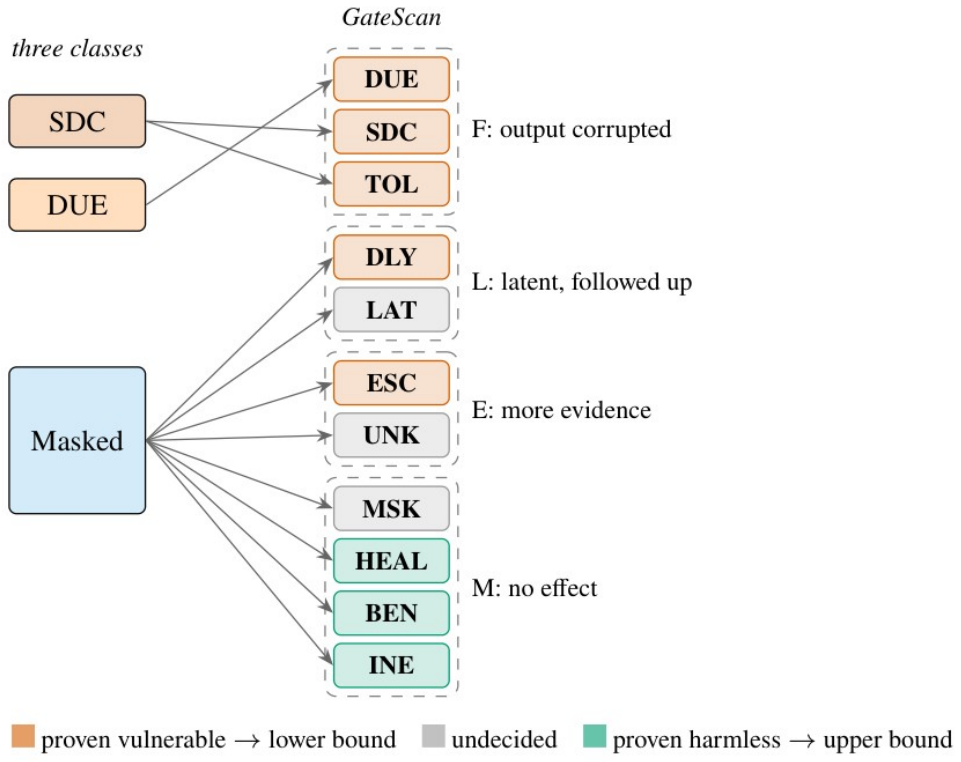
Two simulators and a formal oracle compute every verdict; a wrong agent can only widen the bracket.

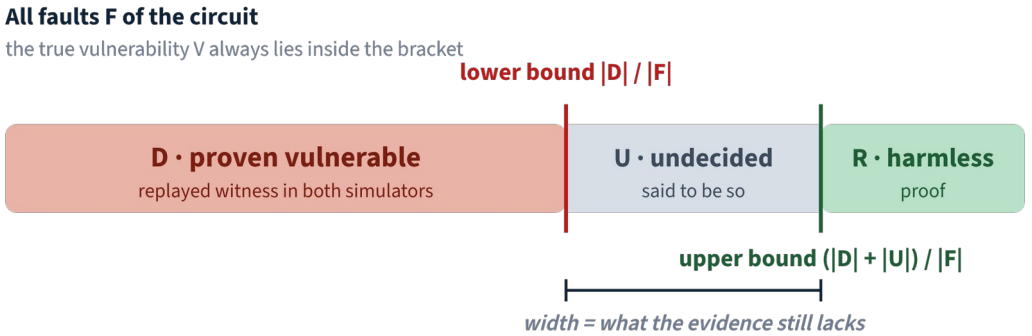
Agents propose, the core decides — **an answer that comes with a guarantee**

Roshani et al., GateScan, IEEE TCAD (draft), Fig. 1 · github.com/gatescan-dev/gatescanMarkIII

From Three Classes to Eleven — and a Certified Bracket

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$$\frac{|D|}{|F|} \leq V \leq \frac{|D| + |U|}{|F|}$$

"Masked" splits into four: ESC (a vulnerability with a replayed witness), MSK, BEN / HEAL and INE.

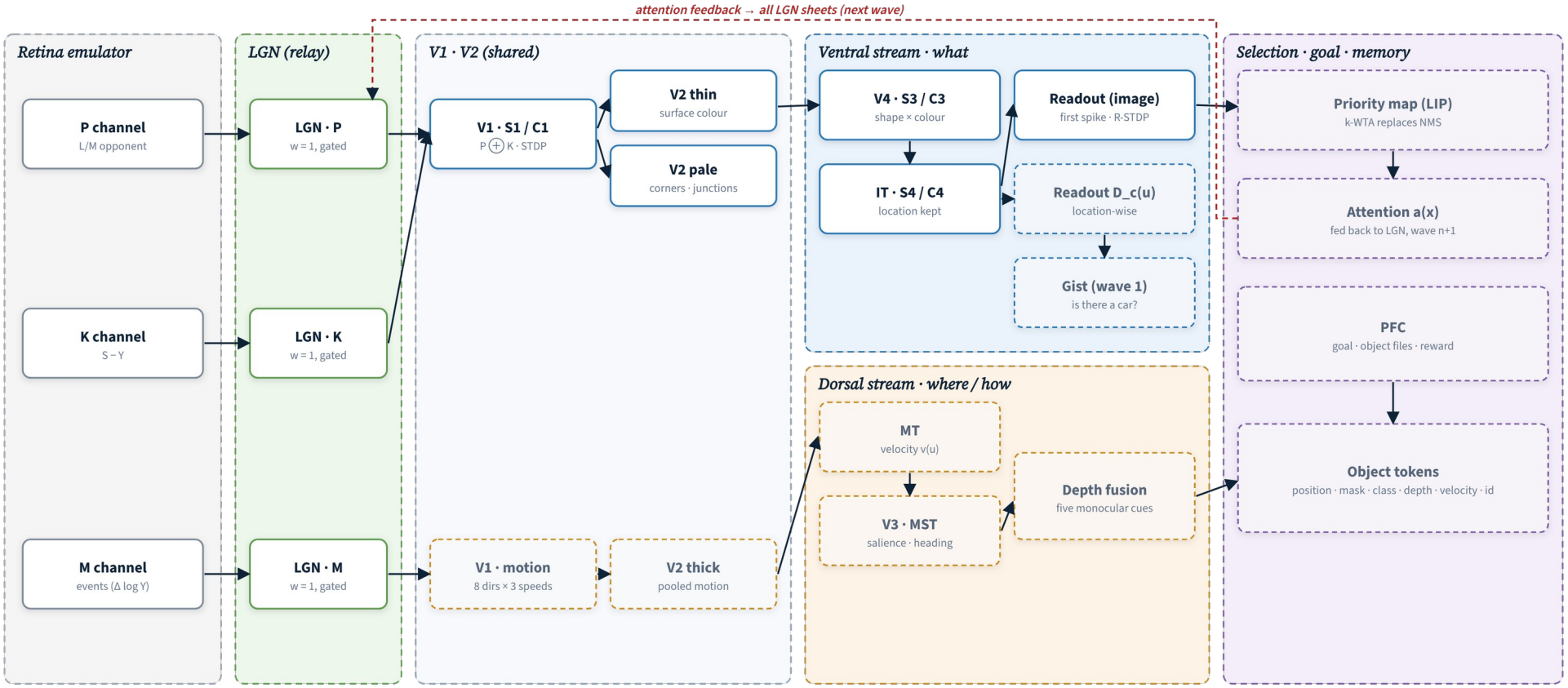
Exact intervals for every sampled share, and an escape certificate that bounds what still hides in U.

"Masked" is not harmless — **we report what is proven, and say what is not**

Roshani et al., GateScan, IEEE TCAD (draft), Sec. III and Fig. 2 · Clopper & Pearson 1934 · hypergeometric and Hoeffding–Serfling intervals

Phase 3 · From One Core to a Brain-Like Vision Network

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The ventral path is built and tested — **motion, attention, memory and depth come next**

BioVision-SNN architecture (draw.io, 2026) · Kheradpisheh et al. 2018 · Mozafari et al. 2018 · Lamme & Roelfsema 2000 (feed-forward sweep vs recurrent waves)

Hardware-Aware by Design

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Retina emulator

P, K and M channels in the log domain; intensity becomes a spike latency — no multipliers.

$$t = T\left(1 - \frac{r}{r_{max}}\right), \qquad r = \frac{\lfloor D \rfloor_+}{\varepsilon + G * Y}$$

Wave-based TTFS

At most one spike per neuron per wave (T = 15 steps); feedback acts in the next wave.

wave 1 | wave 2 | wave 3 | wave 4

Local learning only

Flat STDP in S1–S4, exponential STDP for motion, R-STDP for IT and the read-out; no backprop.

S1–S4	motion	IT · read-out
flat STDP	exp. STDP	R-STDP

Two-sided homeostasis

Lesson from Phase 1: the θ ≥ θ_0 floor silenced S1, so θ now moves both ways — still shifts only.

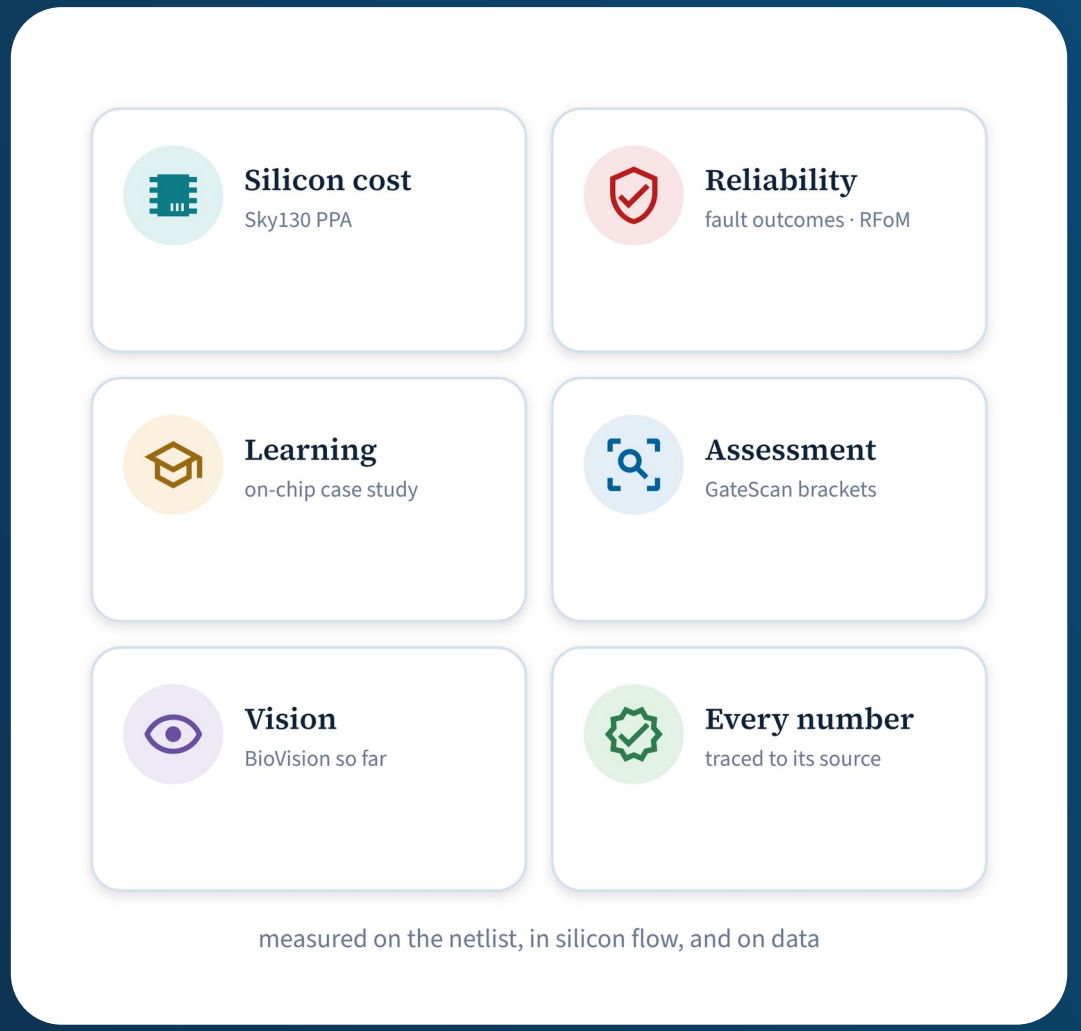
$$\theta_i \leftarrow \theta_i + \eta\,\theta_0\,\frac{z_i - \rho}{\rho}, \qquad \eta = \rho = 2^{-5}$$

Biologically faithful and multiplier-free — **every block maps to shifts, adds and compares**

BioVision-SNN Phase 1 report (notes 16, 26) · Mitchell logarithm · Brader, Senn & Fusi 2007 · Mozafari et al. 2018

04

Results


Silicon cost
Sky130 PPA
Reliability
fault outcomes · RFoM
Learning
on-chip case study
Assessment
GateScan brackets
Vision
BioVision so far
Every number
traced to its source
measured on the netlist, in silicon flow, and on data

Phase 1 · Silicon Cost in SkyWater 130 nm

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	Baseline	Homeostasis	+ Light TMR	+ Restoring Organ	Full TMR
Standard cells	1,772	**2,469**	3,969	**4,093**	6,025
Cell overhead	1.00×	**1.39×**	2.24×	**2.31×**	3.40×
Fmax (MHz)	83.61	**64.92**	64.18	**62.11**	80.00
Total core power (µW)	30.7	**52.8**	88.5	**91.4**	116.8
Energy per update (pJ)	0.367	**0.813**	1.379	**1.472**	1.460

The in-neuron controller is cheap: 697 cells (1.39×) for θ, leak, penalty and brake.

Triplication multiplies everything: 3.40× cells, 3.80× power, 4.12× wire length.

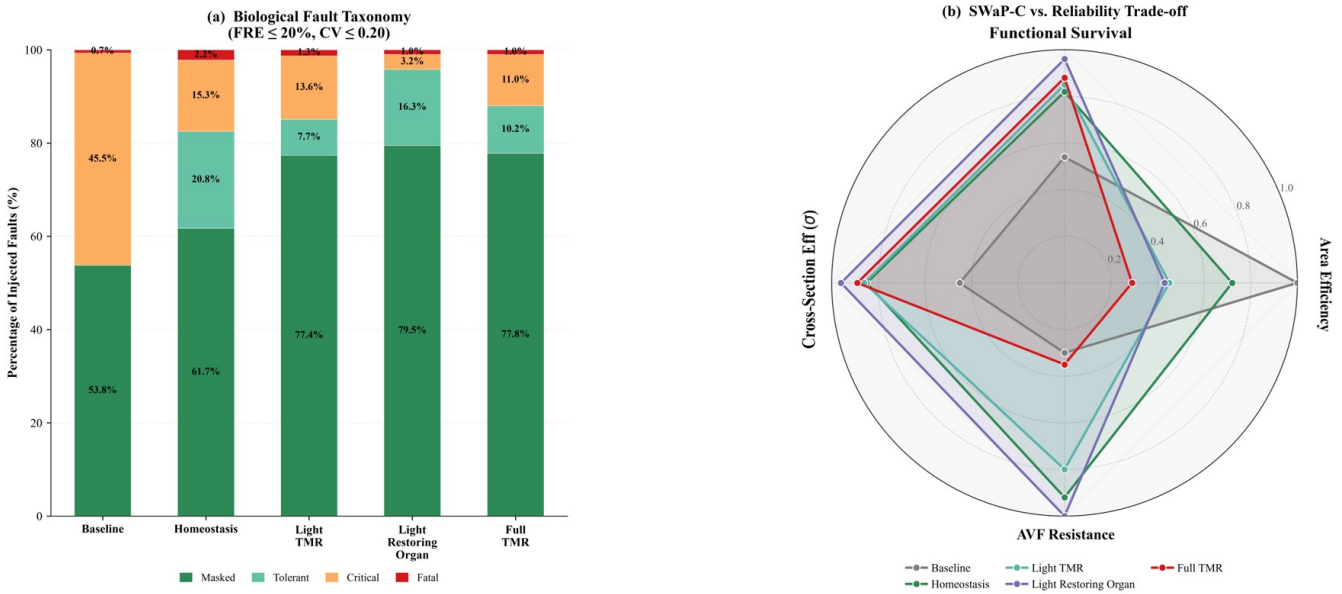
Same energy, different safety: Restoring Organ 1.472 pJ vs full TMR 1.460 pJ per update (next slide).

A fault-tolerant Izhikevich neuron for **under a tenth of a milliwatt**

Roshani et al., TCAS-I (submitted), Table II · OpenLane P&R, OpenSTA sign-off at the typical corner (1.8 V, 25 °C) · same 80,146.87 µm² floorplan for all five designs

Phase 1 · Fault Outcomes on the Netlist

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Dangerous faults (critical + fatal)

Baseline	46.2 %
Homeostasis	17.5 %
+ Light TMR	14.9 %
+ Restoring Organ	4.2 %
Full TMR	12.0 %

Homeostasis alone cuts critical faults 3× and creates a tolerant class (20.8 %). **Restoring Organ: 4.2 % vs 12.0 % for full TMR, with 32 % fewer cells.**

At equal energy per update — **the adaptive core is almost 3× safer than full TMR**

Roshani et al., TCAS-I (submitted), Fig. 9 · exhaustive single stuck-at campaign, both polarities · FRE ≤ 20 % and CV ≤ 0.20

Phase 1 · Reliability per Unit of Silicon

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$$\mathrm{RFoM}_i = \frac{\mathrm{AVF}_{base}/\mathrm{AVF}_i}{N_i/N_{base}}$$

Vulnerability reduction divided by area growth. **A protection pays for its silicon only when RFoM > 1.**

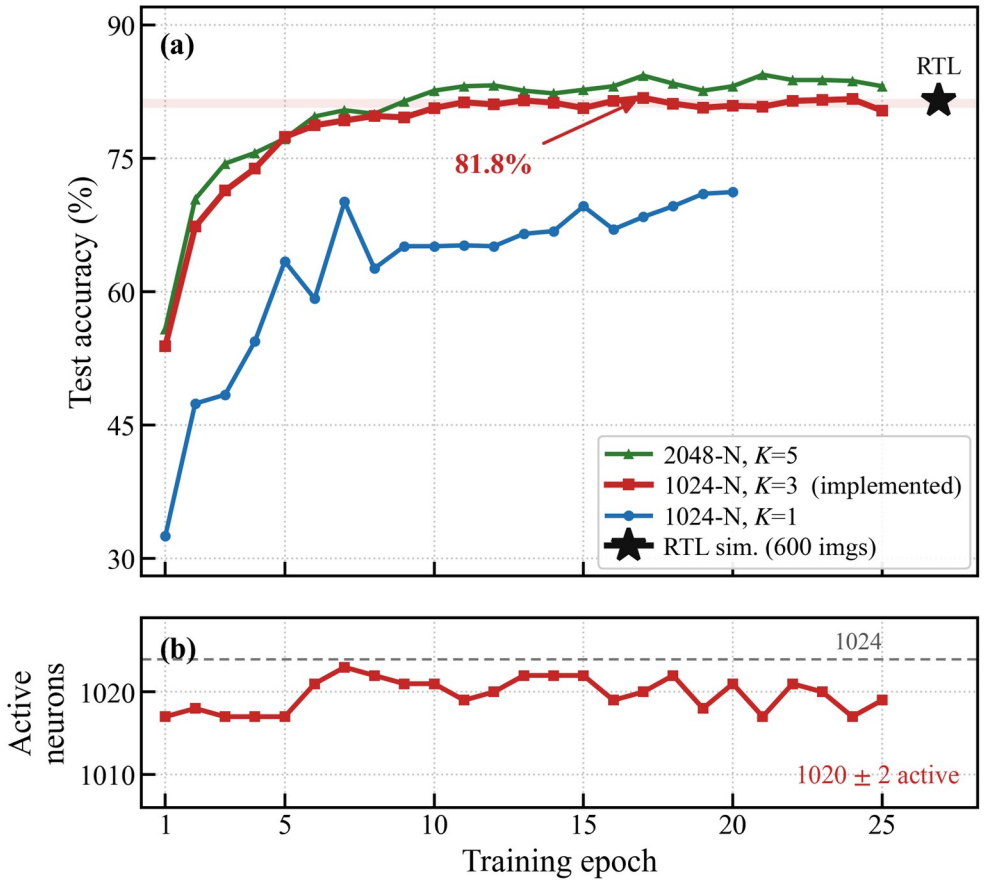
Architecture	Cells	AVF	Vuln. reduction	Area growth	RFoM
Baseline	1,772	0.462	1.00×	1.00×	1.00×
Homeostasis	**2,469**	**0.175**	**2.64×**	**1.39×**	**1.89×**
+ Light TMR	3,969	0.149	3.10×	2.24×	1.38×
+ Restoring Organ	**4,093**	**0.042**	**11.00×**	**2.31×**	**4.76×**
Full TMR	6,025	0.120	3.85×	3.40×	1.13×

Replication is self-limiting — **homeostasis lowers AVF without copying anything**

Roshani et al., TCAS-I (submitted), Table III and Eq. (26) · first-order screening ratio under the single-fault model, not a lifetime

Phase 1 · Learning Survives on the Core

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81.8 %

peak test accuracy (K = 3) · 81.2 ± 0.5 % over the last five epochs

81.5 %

end-to-end RTL simulation on 600 test images — within one standard error

1020 ± 2 / 1024

hidden neurons stay active in all 25 epochs — no death, no dominance

182.8 µW

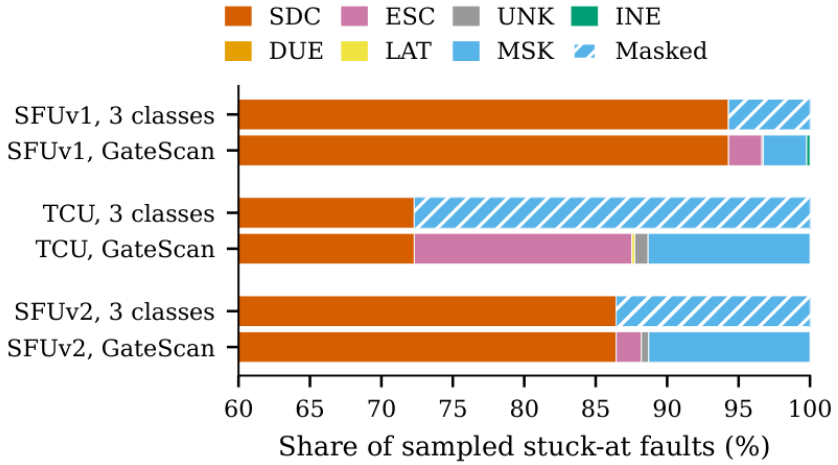
both hardened cores together · 0.177 µW per logical neuron

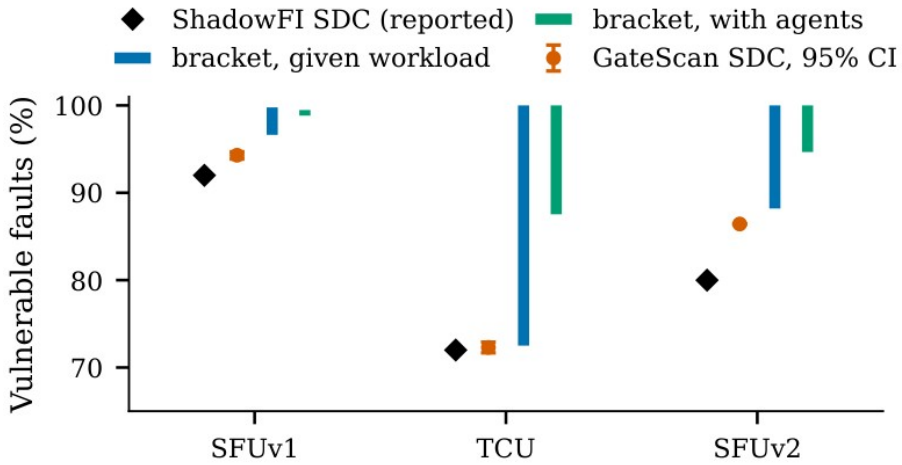
Fault tolerance did not cost learning — **it kept the population alive**

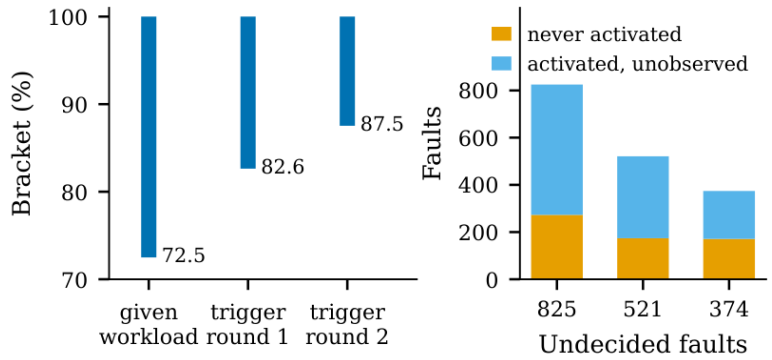
Roshani et al., TCAS-I (submitted), Sec. VII-E and Fig. 10 · 25 epochs on 10,000 MNIST images, evaluated on 2,000 · 256 → 1024 → 10, K = 3

Phase 2 · What the Three-Class View Hides

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13–55 % of “masked” faults are proven vulnerable (ESC)

72.5 → 87.5 % TCU lower bound after two trigger rounds

[3.2, 99.6] bracket on b18, where the point estimate says 3.2 %

0 disagreements in 1,834 two-engine runs · 165/165 gates

A point estimate hides what the campaign does not know — **the bracket shows it**

Roshani et al., GateScan, IEEE TCAD (draft), Table II, Figs. 3, 4 and 6 · ShadowFI benchmarks (SFUv1, TCU, SFUv2) and twenty ITC’99 circuits

Phase 3 · BioVision Results So Far

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PHASE 0 · REPRODUCING THE REFERENCE NETWORKS

Front, dataset	Paper	Ours
RL, Caltech (Mozafari et al.)	98.9 ± 0.4	**98.65 ± 0.49**
SVM, Caltech (Kheradpisheh et al.)	99.1 ± 0.2	**98.96 ± 0.25**
SVM, MNIST	98.4	**98.52 ± 0.06**
SVM, ETH-80, 5/5 split	82.8	**84.25 ± 3.49**

PHASE 1 · COLOUR RETINA + VENTRAL PATH, MULTIPLIER-FREE

89.42 ± 1.18 % RL, Caltech four classes — above the 85 % target

72.86 ± 2.22 % RL, ETH-80 5/5 — training still rising at epoch 80

≈ float multiplier-free retina and learning: 73.48 vs 72.56 % (validation)

4× fewer spikes magno alone ≈ colour on ETH-80 (71.95 vs 71.34 %)

What Phase 1 taught us

- Colour P+K helps on validation (+6.7 %), but the K channel also coded the blue ETH-80 background — found and removed.
- Fixed contrast thresholds are fragile under gamma (72 → 11 %); local contrast normalisation removes it at a 3.7 % cost.
- The H-ASIZH floor $\theta \geq \theta_0$ silenced S1 after STDP converged — replaced by two-sided rate homeostasis.

Local learning, no multipliers — **classification works; detection on COCO is next**

BioVision-SNN Phase 0 and Phase 1 reports (notes 15, 16, 26) · test accuracy, mean ± std over seeds · paper values are best-epoch, ours are selected on validation unless noted

05

Conclusion

What we built
core · campaign · vision

What we learned
adapt, don't replicate

Road ahead
detection · system FT

Outputs
papers · open code

one dependable extreme-edge vision device

Gaps Closed — With Evidence

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G1 No deterministic guarantees	→	C1 One-cycle, Lyapunov-bounded core · 52.8 µW and 0.813 pJ per update
G2 Standards rarely met	→	C2 GateScan: certified brackets and exact intervals, not a percentage
G3 The device is assumed correct	→	C1 Dangerous faults 46.2 % → 4.2 % · RFoM 4.76× vs 1.13× for full TMR
G4 Neuromorphic is not mature	→	C3 BioVision, local learning, no multipliers: 98.5 % MNIST · 89.4 % Caltech-4
G5 No learning in the field	→	C4 On-chip STDP on the hardened core: 81.8 % MNIST, 1020 / 1024 neurons alive

Adapt, don't replicate — **dependability built into the neuron itself**

Numbers: TCAS-I (submitted) Tables II–III and Figs. 9–10 · GateScan TCAD (draft) Table II · BioVision Phase 0–1 reports

The Road Ahead

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BioVision

- Detection: Penn-Fudan, then COCO
- Dorsal stream: motion, depth
- Attention, PFC, object tokens
- Border ownership and masks
- Layer-by-layer fault campaign

Accelerator

- Harden controller, NoC, memory
- Many hardened cores, one SNN
- FPGA prototype
- Tape-out in SkyWater 130 nm

GateScan

- Head-to-head with a commercial tool
- Trigger loop on processors
- Stronger formal oracle
- Million-cell scale

Neuron

- Two-sided fault-tolerant LIF / IF
- Homeostasis 2.0 for learning
- Transient faults (SEU) in the core

Next: from a dependable neuron — **to a dependable vision system**

BioVision project brief, phases 2–5 · GateScan lesson, Ch. 10 · TCAS-I (submitted), Sec. VIII

Research Outputs

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Journal · submitted

H-ASIZH: A Highly Reliable Digital Izhikevich Neuron Core Using Adaptive Threshold Control for Fault Accommodation

A. Roshani, M. Ansari, M. T. Manzuri Shalmani — IEEE Transactions on Circuits and Systems I

Journal · draft

Masked Is Not Harmless: Certified Gate-Level Fault Vulnerability Analysis with LLM Agents That Propose and Engines That Decide

A. Roshani, M. Ansari, M. T. Manzuri Shalmani — IEEE Transactions on Computer-Aided Design

Open source

GLFI fault campaign · GateScan Mark III (core, agents, web service) · BioVision-SNN code and run records

github.com/aliroshanidino/GLFI_fault_campaign · github.com/gatescan-dev/gatescanMarkIII

Reports

BioVision-SNN Phase 0 and Phase 1 technical reports · GateScan lesson (10 chapters) · thesis knowledge base

github.com/aliroshanidino/thesischecking

Two journal papers, open code and open data — **every number in this talk is traceable**

Status as of Aban 1405 / November 2026

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Full bibliography: thesis Chapter 1 (162 references) and the two papers

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Thank you

Questions and discussion

Ali Roshani
Supervisors: Dr. Mohsen Ansari · Dr. Mohammad T. Manzuri Shalmani
CPS Lab · Department of Computer Engineering · Sharif University of Technology
github.com/aliroshanidino